Design and Evaluation of a SIEM-Based Security Monitoring System Using Simulated Attack Scenarios

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Proposal Overview

Problem Summary

Many small organizations lack centralized security monitoring and sufficient visibility into system and network activity, which increases the likelihood that malicious behavior will go undetected until significant damage has occurred. By implementing a Security Information and Event Manager (SIEM) solution, smaller organizations can begin addressing security issues by centralizing log data and improving detection and response to potential security incidents by generating alerts and flags based on logs and system behavior. Without a SIEM, it can be hard to collect, correlate, and analyze security logs; as a result, organizations can struggle to identify, investigate, and respond to threats or incidents. Nevertheless, using a SIEM solution provides improved visibility and faster response times, helping reduce the organization's risk while protecting systems, data, and business operations.

IT Solution

By implementing a Security Information and Event Management (SIEM) system to collect records of activity, also termed logs, from multiple computers and systems. With the collected logs, the SIEM can analyze them and compare them against normal network behavior to identify any signs of unusual or suspicious activity. The SIEM centralizes information, provides the company with a single view of network activity, and generates alerts when there is anything concerning or worth a deeper look. The purpose of the SIEM in this project is to monitor simulated systems for a simulated company and evaluate its ability to identify and detect

potential security incidents, thereby demonstrating the importance of a SIEM system for a company and how it can improve response time, efficiency, and incident-detection accuracy.

Implementation Plan

The first phase of the project is to create a virtual lab environment that simulates a small organization's network. The second phase involves installing the SIEM system and setting up to receive logs from multiple computers. Then, for phase three, I will simulate regular system activity across multiple computers, which will be monitored and observed to help create a baseline of expected network behavior for the SIEM to use as a comparison to other activity. The fourth phase is where I will simulate suspicious activity on the computers and network, now that there is a baseline to compare these logs to. The suspicious activities will test whether the SIEM can properly detect and alert on these behaviors, bringing them to light. Lastly, the results will be analyzed and documented to evaluate how effectively the SIEM monitored the logs and the benefits to the company. The virtual environment is used for resource limitations and to contain the simulated network, reducing risk when simulating attacks. The attacks themselves are not what this project focuses on, but rather the evaluation of a security monitoring system within an organization. Thus, the attacks will not be as advanced as trying to hack my way into the network via an attack machine; instead, they will assume the attacker already has the necessary information on the endpoints to attempt logins, gain persistence, move laterally, gain permissions, and alter data on the systems. These attacks are very real risks when organizations are under attack, but the overall goal is to assess how effective the SIEM is and what it catches or misses.

Review of Other Work

Summary of Four Works

Work 1: Systematic review of SIEM technology: SIEM-SC birth.

This article provides a broad review of Security Information and Event Management (SIEM) systems and how they have changed and evolved. SIEM systems collect logs from many devices across an organization and analyze them to detect suspicious behavior, as the author explains. This study shines when discussing how SIEM technology has evolved from basic log storage to real-time monitoring and alerting. Real-time monitoring enables the organization to respond more quickly to security incidents. Though it also highlights some conflicts or challenges, including high resource requirements and the need for trained personnel. This study concludes that SIEM systems have been and remain essential for enhancing security visibility and reducing incident response time.

Work 2: A Systematic Review of Cyber Threat Intelligence: The Effectiveness of Technologies, Strategies, and Collaborations in Combating Modern Threats.

The second study I have chosen to support my project examines cyber threat intelligence. The study focuses on the process of collecting and analyzing information on potential cyber threats to help organizations respond to attacks and prepare for them. By analyzing large volumes of security data or logs, organizations can identify patterns that may indicate malicious behavior. The data author should come from multiple systems to improve accuracy. It is through these built-in patterns that detection will be improved and the risk of malicious activity going unnoticed will be reduced. While the author discusses advanced techniques, the key points of the

study are that better analysis of security data will significantly improve an organization's ability to detect and respond to attacks.

Work 3: Deployment of Honeypot and SIEM Tools for Cyber Security Education Model in UITM.

This study describes the deployment of a cybersecurity laboratory that uses a SIEM system to help students understand how attacks are analyzed and detected. Simulated environments allow users to observe and study real security events without risking vulnerabilities or threats to production systems or host machines. The author has shown in this study that hands-on practice with a SIEM to analyze activity on a system or network greatly improves understanding of how security monitoring works and how suspicious or malicious behavior is identified and alerted.

Work 4: Cybersecurity on a budget: Evaluating security and performance of open-source SIEM solutions for SMEs.

The purpose of this study is to demonstrate the effectiveness of open-source SIEM solutions for organizations, both small and medium, with limited budgets or staff. Many organizations struggle to monitor all their systems because so many logs arrive at once and are scattered across systems. The author describes how a SIEM system can help with this issue, acting as a centralized dashboard to collect and organize logs while still alerting the organization to any suspicious behavior. With open-source SIEM tools, these organizations can still get security monitoring for their systems without the cost of commercial SIEM products. It is the price and

power that make open-source SIEM products such a viable solution, primarily since they can deliver as much power as most commercial products when used and configured correctly.

Relation of Works to Proposal Design

The reviewed works for this project collectively support the design and approach of my project, as they establish the importance of creating a centralized security monitoring tool, improving data collection and analysis, and using a controlled testing environment. Prior research shows that organizations struggle to detect security incidents or malicious behavior on their systems and networks because logs are spread across a wide range of systems, are unorganized, and are challenging to keep up with and review promptly, thereby slowing response times (López Velásquez et al., 2023). Based on the information provided, a centralized monitoring system must be implemented to collect and analyze data, flag or alert the organization to any risks and/or suspicious behavior, and support the company's overall security posture.

The research study on cybersecurity threat intelligence highlights the importance of analyzing collected data holistically rather than focusing on individual events to identify patterns indicative of malicious behavior (Santos et al., 2025). The ability to analyze and identify patterns across various data sources will help detect these behaviors before it is too late, giving the organization more time to react when a security incident is in progress. This study helps reinforce the importance of how collected logs must be analyzed and then used to cross-compare to current logs, so that way alerts can be generated when something is out of the ordinary and

may need further inspection rather than just storing the logs on their own and seeing security incidents as they happen thus putting the organization in a proactive approach rather than purely responsive.

The third study I chose to help support my project examines how simulated systems provide a safe and effective way to practice and learn about security tools without risking real organizational data in the area of cybersecurity education and safe lab environments (Mohd Ariffin et al., 2022). This research demonstrates that simulated malicious behavior within a virtualized environment can still be analyzed and collected, just as in any other SIEM, and that this analysis can inform how monitoring systems respond to these threats. My project's foundation lies in using a virtual environment to mimic a real organization's environment, without actually using an organization's environment, allowing me to conduct simulated attacks and activities ethically, all for the primary purpose of evaluating and demonstrating the use of monitoring systems and their detection capabilities.

My final research study for this project discusses how an open-source monitoring system can be both beneficial and as effective as other commercial products (Manzoor et al., 2024). Support for open-source SIEM systems reinforces the relevance of this project's scope and gives credibility to the SIEM tool selected for this project, even though it is open source. The focus is on demonstrating practical security monitoring in a small organizational environment. Many organizations that lack the budget or resources for commercial SIEMs commonly use open-source SIEMs to make up for this shortfall (Manzoor et al., 2024).

All together, these works help build the foundation for my project's design by validating the need for a centralized monitoring system, the use of data analysis for threat detection, the use of a virtualization environment, and the suitability and relevance of an open-source SIEM in a small organizational setting with limited resources. This project builds on these ideas to develop a practical implementation and evaluation of a SIEM-based monitoring system.

Project Rationale

The reason this project was selected is that many organizations, especially smaller ones that have limited resources and visibility, struggle to detect security misconfigurations, suspicious behavior on their systems, and security incidents promptly. These faults cause them to respond slowly to these situations, leading to information being exfiltrated and stolen. It is these organizations that we trust with our own data that need to be more proactive in responding to cyberattacks and related risks. Security monitoring and incident detection are essential components of any modern cybersecurity operation. This project aligns with future career goals related to security operations and incident response.

Current Project Environment

The project will be conducted within a virtual lab environment hosted on a personal computer using a virtualization manager. The environment will consist of one system responsible for centralized monitoring and analysis. In contrast, the other two systems will represent typical organizational computers found around the office and used by staff. All systems within the environment will be connected via a private virtual network to ensure isolation from external

networks. The controlled environment allows for safe testing and proper evaluation of the proposed solution.

Methodology

This project will follow a design-and-evaluation methodology, in which the monitoring system will be designed and implemented in the virtual environment on the central server. Then, regular system activity will be simulated on the other two endpoints, acting as employee host machines, to build a baseline. Once a solid baseline is established, attacks and suspicious behavior will be simulated on the host machines to test how the SIEM responds. The SIEM's alerts and log records will then be analyzed to determine whether the SIEM successfully, accurately, and effectively identified potential security issues and whether they were identified promptly. The last step is to evaluate and document the findings to determine whether the SIEM was effective in its purpose of improving response time, organizing and capturing logs, and alerting the organization to suspicious behavior, thereby providing further evidence of how a security monitoring system can help companies improve their security posture.

Project Goals, Objectives, and Deliverables

The purpose of this project is to design, create, implement, simulate, and evaluate a SIEM solution that improves visibility into an organization's architecture and system activity logs and helps detect malicious behavior on the network. To achieve this goal, the project has been built on a foundation of research and clear goals. Each goal represents an outcome that the project aims to achieve, whereas the objectives outline the steps to achieve these outcomes. Lastly, the deliverables will provide tangible evidence that the goals have been successfully achieved.

Goals, Objectives, and Deliverables Descriptions

Goal 1: Designing a SIEM that will provide visibility into system activity.

Goal one focuses on the design of a centralized monitoring system that provides a clear view of activity across multiple computers in a small organization environment consisting of one server and two endpoint machines. This goal aims to address the problem of scattered, unorganized log data, which can lead to many logs not being reviewed adequately, efficiently, and on time, or, in some cases, to their complete oversight. The SIEM helps achieve this goal by providing a centralized location for collecting and analyzing all log data. To achieve this goal, a virtual lab environment will be created to represent the small organization, and a SIEM will be installed within the network to collect and analyze data. An architecture or network diagram will be included to map how data is moving throughout the environment, to describe the environment itself, and, lastly, to document how the monitoring system will be set up to collect log data from the entire network.

Goal 2: Evaluating the SIEM's ability to detect malicious activity and overall effectiveness.

The second goal for the project is to evaluate the effectiveness of the SIEM in identifying unusual or potentially harmful network behavior and ensuring it does not improperly flag or alert on normal operating behavior. To fulfill this goal, a baseline will be created first by observing regular system activity and logs, providing the SIEM with something to build from and compare against the malicious behavior that will be simulated later. Once that baseline is documented, simulated attacks and suspicious activity will be performed in the lab environment on the two endpoint machines. The simulated harmful activity is intended to be designed and represented as common security concerns, such as repeated login failures that could signal a brute-force attack, and as actions that build on the MITRE ATT&CK Framework to simulate an attack. The SIEM's alerts will be observed and documented to determine whether the simulated attacks were indeed detected or not, along with how accurate the data is on the attacks, such as how the data can be used to create new security safeguards, and whether there are false positives or false negatives, and if there are, then how often do they occur. The deliverables for this goal are to create and document the baseline activities in a summary, descriptions of the simulated attack behaviors, and screenshots or descriptions of the alerts captured and generated by the SIEM.

Goal 3: Data analysis and demonstrate how incidents can be identified and investigated.

The third goal of the project is to analyze the alerts and logs generated by the SIEM during normal operations and simulated activity. The purpose of analyzing the data is to demonstrate how potential security incidents can be identified and investigated, sometimes even before attackers can gain access to the machines. Goal three emphasizes understanding and interpreting

the logs rather than just configuring the SIEM. The alerts generated will be reviewed, and the associated logs will be examined to explain why the behavior was flagged. This analysis will then be used to create an example incident walkthrough demonstrating how detected activity could be investigated in a real organizational setting. Deliverables for this goal would include creating tables to analyze alerts and their meanings, creating an incident walkthrough to show how an organization can investigate the behavior, and providing written explanations of potential security risks and ways to help remediate them.

Goals, Objectives, and Deliverables Table

	Goal	**Supporting Objectives**	**Deliverables Enabling the Project Objectives**
1	Designing a SIEM that will provide visibility into system activity.	Create a virtual environment representing a small organization.	System Archetecture Diagram
			Network Diagram
			Description of Virtual Environment
		Install a centralized monitoring system to collect activity records from multiple computers.	Documentation of monitoring system setup
2	Evaluating the SIEM's ability to detect malicious activity and overall effectiveness.	Establish normal behavior as a baseline.	Create a baseline activity summary
			Document monitoring dashboard of normal behavior
		Perform simulated attack behavior in the controlled environment.	Descriptions of simulated attacks
			Connect simulated attacks to phases in MITRE ATT&CK Framework
		Observe alerts and evaluate SIEM system.	Explanation of what caused each alert
			Evaluation of false-positives/false-negatives
			Evaluation of remediation steps and new security safeguards
3	Data analysis and	Interpret the potential security risk of detected behaviors.	Summary of potential impact on organization

	demonstrate how incidents can be identified and investigated.		Explanation of each alert and their risk
		Demonstrate an example incident investigation.	Written incident walkthrough
			Incident summary explanation for management

Project Timeline with Milestones

Milestone	**Duration (hours or days)**	**Projected Start Date**	**Anticipated End Date**
Virtual Environment setup and configurations	1 Day	Feb. 12, 2026	Feb. 13, 2026
SIEM installation and system configuration	2 Days	Feb. 12, 2026	Feb. 14, 2026
Baseline activity observation and documentation	1 Day	Feb. 14, 2026	Feb. 15, 2026
Simulated attacks and alert generation from SIEM	1 Day	Feb. 15, 2026	Feb. 16, 2026
Alert analysis and incident walkthrough development	1 Day	Feb. 16, 2026	Feb. 17, 2026
Drafting final capstone and review, revise, submit	7 Days	Feb. 17, 2026	Feb. 24, 2026

Outcome

The expected outcome of this project is a functional, efficient security monitoring system that provides better visibility into system activity and demonstrates the ability to detect behavior

outside normal operating baselines in a controlled virtual lab environment. By centralizing logs from multiple systems, the SIEM will generate alerts that may indicate potential security issues, along with supporting structured investigations to determine the effectiveness of those alerts. The project will produce evidence documenting how simulated attack activities can be identified, analyzed, and communicated to management through centralized monitoring.

The project will also include a comprehensive written report that explains the design, implementation, and evaluation of the SIEM in plain language, using the findings to demonstrate how centralized monitoring can improve detection and response capabilities while reducing risk within the organization. Overall, the project will illustrate the value and practicality of security monitoring systems for small- to medium-sized organizations, as well as how improved visibility into internal systems and networks supports decision-making and accelerates incident response plans.

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